

Continuous Relaxation for Graph Isomorphism Detection

1 Problem Formulation

Given two simple, undirected graphs $G_1 = (V, E_1)$ and $G_2 = (V, E_2)$ on n vertices, we seek to determine whether they are isomorphic. Two graphs are isomorphic if there exists a permutation $\pi : V \rightarrow V$ such that $(u, v) \in E_1 \iff (\pi(u), \pi(v)) \in E_2$.

Let $\mathbf{A}, \mathbf{B} \in \{0, 1\}^{n \times n}$ denote the adjacency matrices of G_1 and G_2 respectively. A permutation π can be represented as a permutation matrix $\mathbf{P} \in \{0, 1\}^{n \times n}$ where $P_{ij} = 1$ iff $\pi(i) = j$. The graphs are isomorphic iff there exists a permutation matrix $\mathbf{P}$ such that:

$$\mathbf{P}^\top \mathbf{A} \mathbf{P} = \mathbf{B} \quad (1)$$

Since searching over all $n!$ permutations is computationally intractable, we relax the problem to a continuous optimization over *doubly stochastic matrices*.

2 Doubly Stochastic Relaxation

A matrix $\mathbf{X} \in \mathbb{R}^{n \times n}$ is **doubly stochastic** if:

$$X_{ij} \geq 0, \quad \sum_{j=1}^n X_{ij} = 1 \quad \forall i, \quad \sum_{i=1}^n X_{ij} = 1 \quad \forall j \quad (2)$$

The set of doubly stochastic matrices forms the *Birkhoff polytope* $\mathcal{B}_n$, whose vertices are exactly the $n!$ permutation matrices (Birkhoff–von Neumann theorem). By optimizing over $\mathcal{B}_n$, we obtain a convex relaxation of the combinatorial problem.

3 Objective Function

We formulate a quadratic program (QP) that penalizes structural mismatch between the graphs under a soft permutation $\mathbf{X}$.

3.1 Degree Mismatch (Linear Term)

Define the degree vectors $\mathbf{d}_i^A = \sum_k A_{ik}$ and $\mathbf{d}_j^B = \sum_k B_{jk}$. The cost matrix:

$$C_{ij} = |d_i^A - d_j^B| \quad (3)$$

penalizes mapping node i in G_1 to node j in G_2 if their degrees differ. The linear objective term is:

$$\mathcal{L}(\mathbf{X}) = \sum_{i=1}^n \sum_{j=1}^n C_{ij} X_{ij} = \langle \mathbf{C}, \mathbf{X} \rangle_F \quad (4)$$

3.2 Adjacency Mismatch (Quadratic Term)

For a true permutation matrix $\mathbf{P}$, the transformed adjacency would be $\mathbf{P}^\top \mathbf{A} \mathbf{P}$. We penalize the discrepancy between $\mathbf{A}$ transformed by $\mathbf{X}$ and $\mathbf{B}$ transformed by $\mathbf{X}$:

$$\mathcal{Q}(\mathbf{X}) = \sum_{p=1}^n \sum_{q=1}^n \left(\sum_{k=1}^n A_{pk} X_{kq} - \sum_{k=1}^n X_{pk} B_{kq} \right)^2 \quad (5)$$

This can be written in matrix form. Let $\mathbf{R} = \mathbf{A}\mathbf{X} - \mathbf{X}\mathbf{B}$, then:

$$\mathcal{Q}(\mathbf{X}) = \|\mathbf{A}\mathbf{X} - \mathbf{X}\mathbf{B}\|_F^2 = \sum_{p,q} R_{pq}^2 \quad (6)$$

When $\mathbf{X} = \mathbf{P}$ is a valid isomorphism, we have $\mathbf{A}\mathbf{P} = \mathbf{P}\mathbf{B}$, hence $\mathcal{Q}(\mathbf{P}) = 0$.

4 Quadratic Program

The complete optimization problem is:

$$\begin{array}{ll} Z^* = \min_{\mathbf{X}} & \mathcal{L}(\mathbf{X}) + \mathcal{Q}(\mathbf{X}) \\ \text{s.t.} & \sum_{j=1}^n X_{ij} = 1 \quad \forall i \in \{1, \dots, n\} \\ & \sum_{i=1}^n X_{ij} = 1 \quad \forall j \in \{1, \dots, n\} \\ & X_{ij} \geq 0 \quad \forall i, j \end{array} \quad (7)$$

This is a convex QP (the quadratic term is a sum of squares, hence convex) and can be solved efficiently using standard solvers such as Gurobi.

5 Isomorphism Index

The optimal objective value Z^* measures the “distance” from isomorphism:

- $Z^* = 0$: The graphs are isomorphic (a perfect permutation exists)
- $Z^* > 0$: Structural mismatch; larger values indicate greater dissimilarity

We define a normalized **Isomorphism Index** $I \in (0, 1]$:

$$I = \exp(-\lambda \cdot Z^*) \quad (8)$$

where $\lambda > 0$ is a scaling parameter (default $\lambda = 0.1$). This maps the unbounded objective to a bounded similarity score:

- $I \approx 1$: Graphs are (likely) isomorphic
- $I \ll 1$: Graphs are structurally dissimilar

6 Interpretation of the Solution

The optimal $\mathbf{X}^*$ is a doubly stochastic matrix representing a “soft” node correspondence:

- If $\mathbf{X}^*$ is a permutation matrix (binary), it directly encodes the isomorphism mapping
- If $\mathbf{X}^*$ has fractional entries, the relaxation is not tight—this typically occurs for non-isomorphic graphs where no perfect matching exists

7 Complexity

The QP has n^2 variables and $2n$ equality constraints. Modern interior-point solvers achieve polynomial complexity in practice, making this approach tractable for moderate graph sizes ($n \lesssim 100$).